

```
import cv2
image = cv2.imread('image.jpg')
cv2.imshow('Display Image', image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

#### Key features:

- Extensive tools for image and video analysis.
- Support for multiple programming languages.
- Optimised for real-time applications.

#### Use cases:

- Face detection and recognition
- Object tracking in videos

## 8 NLTK (Natural Language Toolkit)

NLTK is a leading platform for building Python programs to work with human language data.

**Example:** Tokenizing a sentence:

```
import nltk
nltk.download('punkt')

from nltk.tokenize import word_tokenize
sentence = "Machine learning is amazing!"
print(word_tokenize(sentence))
```

#### Key features:

- Tools for tokenization, stemming, and tagging.
- Built-in corpora and lexical resources.
- Suitable for both beginners and experts.

#### Use cases:

- Sentiment analysis
- Text summarisation

## 9 RAPIDS

RAPIDS is an open source suite of software libraries and APIs for executing end-to-end data science pipelines entirely on GPUs.

**Example:** Accelerated data processing using cuDF:

```
import cudf

data = cudf.DataFrame({'a': [1, 2, 3], 'b': [4, 5, 6]})
print(data.describe())
```

#### Key features:

- GPU-accelerated data processing.
- Seamless integration with PyData ecosystem.
- High-performance ML algorithms.

#### Use cases:

- Large-scale data analysis
- Recommender systems

## 10 DVC (Data Version Control)

DVC is an open source version control system for machine learning projects, enabling reproducibility and collaboration.

**Example:** Versioning a dataset:

```
dvc init
dvc add data.csv
git commit -m "Add dataset"
```

#### Key features:

- Versioning of datasets and models.
- Integration with Git for seamless workflow.
- Support for cloud and local storage.

#### Use cases:

- Managing ML experiments
- Collaborative ML development

Open source tools have become the backbone of modern AI and ML development. By leveraging these tools, developers and researchers can build sophisticated models, accelerate innovation, and contribute to a growing community. Whether you're a beginner or an experienced professional, exploring these tools will help you stay at the forefront of AI and ML advancements. **END** 

### References

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